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5TH CSE 'H'

Lab-7-Fuzz-Testing

#46 photo_print_order

Run the buggy application photo_print_order.py in the repo assigned to you (under Lab7):

(Execution of the Original Buggy Application)

```
C:\Users\srish\Downloads\L6_L7_46_photo_print_order-main\L6_L7_46_photo_prin
t_order-main\Lab 7>python photo_print_order.py
==== Running parse_order ====
Enter print size:copies (e.g. 4x6:25): 4x6:25
Size and Copies: ('4X6', 25)

==== Running file_size_average ====
Enter file sizes in MB as CSV (e.g. 4,8,12): 4,8,12
Average File Size: 8

==== Running payable_amount ====
Enter amount|discount (e.g. 450|50): 450|50
Payable Amount: 481.0

C:\Users\srish\Downloads\L6_L7_46_photo_print_order-main\L6_L7_46_photo_prin
t_order-main\Lab 7>
```

Complete the code in test_processor.py using hypothesis (Hypothesis Test Results Showing Failures Before Fixing)

```
C:\Users\srish\Downloads\L6_L7_46_photo_print_order-main\L6_L7_46_photo_print_order-main\Lab 7>python -m pytest -v test_photo_print_order.py
============================= test session starts =============================
platform win32 -- Python 3.13.5, pytest-9.2.0, pluggy-1.7.0 -- C:\Python313\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\srish\Downloads\L6_L7_46_photo_print_order-main\L6_L7_46_photo_print_order-main\Lab 7
plugins: anyio-4.10.0, hypothesis-6.170.15
collected 3 items

test_photo_print_order.py::test_parse_order_no_crash FAILED          [ 33%]
test_photo_print_order.py::test_file_size_average_no_crash FAILED    [ 66%]
test_photo_print_order.py::test_payable_amount_no_crash FAILED       [100%]

================================== FAILURES ===================================
_____________________ test_parse_order_no_crash _____________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\srish\Downloads\L6_L7_46_photo_print_order-main\L6_L7_46_photo_print_order-main\Lab 7\test_photo_print_order.py", line 11, in test_pars
e_order_no_crash
  |     def test_parse_order_no_crash(s):
  |                ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\srish\Downloads\L6_L7_46_photo_print_order-main\L6_L7_46_photo_print_order-main\Lab 7\test_photo_print_order.py", line 12, in test_pa
rse_order_no_crash
    |     parse_order(s)
    |     ~~~~~~~~~~~~^^^
    |   File "C:\Users\srish\Downloads\L6_L7_46_photo_print_order-main\L6_L7_46_photo_print_order-main\Lab 7\photo_print_order.py", line 3, in parse_order
    |     return size.strip().upper(), int(copies)
    |                                  ~~~^^^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    | Failing test case: test_parse_order_no_crash(
    |     s=':',
    | )
    +---------------- 2 ----------------
```

```
    | Traceback (most recent call last):
    |   File "C:\Users\srish\Downloads\L6_L7_46_photo_print_order-main\L6_L7_46_photo_print_order-main\Lab 7\test_photo_print_order.py", line 12, in test_pa
rse_order_no_crash
    |     parse_order(s)
    |     ~~~~~~~~~~~~^^^
    |   File "C:\Users\srish\Downloads\L6_L7_46_photo_print_order-main\L6_L7_46_photo_print_order-main\Lab 7\photo_print_order.py", line 2, in parse_order
    |     size, copies = text.split(':')
    |     ^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    | Failing test case: test_parse_order_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\srish\Downloads\L6_L7_46_photo_print_order-main\L6_L7_46_photo_print_order-main\Lab 7\test_photo_print_order.py", line 12, in test_pa
rse_order_no_crash
    |     parse_order(s)
    |     ~~~~~~~~~~~~^^^
    |   File "C:\Users\srish\Downloads\L6_L7_46_photo_print_order-main\L6_L7_46_photo_print_order-main\Lab 7\photo_print_order.py", line 2, in parse_order
    |     size, copies = text.split(':')
    |                    ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    | Failing test case: test_parse_order_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
_____________________ test_file_size_average_no_crash ______________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\srish\Downloads\L6_L7_46_photo_print_order-main\L6_L7_46_photo_print_order-main\Lab 7\test_photo_print_order.py", line 17, in test_file
_size_average_no_crash
  |     def test_file_size_average_no_crash(s):
  |                                         ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\srish\Downloads\L6_L7_46_photo_print_order-main\L6_L7_46_photo_print_order-main\Lab 7\test_photo_print_order.py", line 18, in test_fi
le_size_average_no_crash
    |     file_size_average(s)
    |     ~~~~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\srish\Downloads\L6_L7_46_photo_print_order-main\L6_L7_46_photo_print_order-main\Lab 7\photo_print_order.py", line 6, in file_size_ave
rage
    |     values = [int(r) for r in sizes_csv.split(',')]
    |                  ~~~^^^
    | ValueError: invalid literal for int() with base 10: ''
    | Failing test case: test_file_size_average_no_crash(
    |     s='',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\srish\Downloads\L6_L7_46_photo_print_order-main\L6_L7_46_photo_print_order-main\Lab 7\test_photo_print_order.py", line 18, in test_fi
le_size_average_no_crash
    |     file_size_average(s)
    |     ~~~~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\srish\Downloads\L6_L7_46_photo_print_order-main\L6_L7_46_photo_print_order-main\Lab 7\photo_print_order.py", line 6, in file_size_ave
rage
    |     values = [int(r) for r in sizes_csv.split(',')]
    |                              ^^^^^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    | Failing test case: test_file_size_average_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
_____________________ test_payable_amount_no_crash _____________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\srish\Downloads\L6_L7_46_photo_print_order-main\L6_L7_46_photo_print_order-main\Lab 7\test_photo_print_order.py", line 23, in test_paya
ble_amount_no_crash
  |     def test_payable_amount_no_crash(s):
  |                                       ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\srish\Downloads\L6_L7_46_photo_print_order-main\L6_L7_46_photo_print_order-main\Lab 7\test_photo_print_order.py", line 24, in test_pa
yable_amount_no_crash
    |     payable_amount(s)
    |     ~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\srish\Downloads\L6_L7_46_photo_print_order-main\L6_L7_46_photo_print_order-main\Lab 7\photo_print_order.py", line 11, in payable_amou
nt
    |     total = float(amount) * 1.18 - float(discount)
    |             ~~~~~^^^^^^^^
    | ValueError: could not convert string to float: ''
    | Failing test case: test_payable_amount_no_crash(
    |     s='|',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\srish\Downloads\L6_L7_46_photo_print_order-main\L6_L7_46_photo_print_order-main\Lab 7\test_photo_print_order.py", line 24, in test_pa
yable_amount_no_crash
    |     payable_amount(s)
    |     ~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\srish\Downloads\L6_L7_46_photo_print_order-main\L6_L7_46_photo_print_order-main\Lab 7\photo_print_order.py", line 10, in payable_amou
nt
    |     amount, discount = text.split('|')
    |     ^^^^^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    | Failing test case: test_payable_amount_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\srish\Downloads\L6_L7_46_photo_print_order-main\L6_L7_46_photo_print_order-main\Lab 7\test_photo_print_order.py", line 24, in test_pa
yable_amount_no_crash
    |     payable_amount(s)
    |     ~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\srish\Downloads\L6_L7_46_photo_print_order-main\L6_L7_46_photo_print_order-main\Lab 7\photo_print_order.py", line 10, in payable_amou
nt
    |     amount, discount = text.split('|')
    |                        ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    | Failing test case: test_payable_amount_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
========================= short test summary info =========================
FAILED test_photo_print_order.py::test_parse_order_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_photo_print_order.py::test_file_size_average_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_photo_print_order.py::test_payable_amount_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
========================= 3 failed in 18.62s ==========================
```

Run the fixed photo_print_order.py (Execution of the Fixed Application)

```
                  python photo_print_order.py
==== Running parse_order ====7_46_photo_print_order-main\L6_L7_46_photo_prin
t_order-main\Lab 7>
Enter print size:copies (e.g. 4x6:25): 4x6:25
Size and Copies: ('4X6', 25)

==== Running file_size_average ====
Enter file sizes in MB as CSV (e.g. 4,8,12): 4,8,12
Average File Size: 8

==== Running payable_amount ====
Enter amount|discount (e.g. 450|50): 450|50
Payable Amount: 481.0

C:\Users\srish\Downloads\L6_L7_46_photo_print_order-main\L6_L7_46_photo_prin
t_order-main\Lab 7>
```

Re-Run the test_processor.py wrt to fixed processor and take the ss of the output (Successful Hypothesis Test Results After Fixing)

```
C:\Users\srish\Downloads\L6_L7_46_photo_print_order-main\L6_L7_46_photo_prin
t_order-main\Lab 7>python -m pytest -v test_photo_print_order.py
============================= test session starts =============================
platform win32 -- Python 3.13.5, pytest-9.2.0, pluggy-1.7.0 -- C:\Python313\
python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\srish\Downloads\L6_L7_46_photo_print_order-main\L6_L7_46_p
hoto_print_order-main\Lab 7
plugins: anyio-4.10.0, hypothesis-6.170.15
collected 3 items

test_photo_print_order.py::test_parse_order_no_crash PASSED              [ 33%]
test_photo_print_order.py::test_file_size_average_no_crash PASSED        [ 66%]
test_photo_print_order.py::test_payable_amount_no_crash PASSED           [100%]

============================== 3 passed in 9.15s ==============================

C:\Users\srish\Downloads\L6_L7_46_photo_print_order-main\L6_L7_46_photo_prin
t_order-main\Lab 7>
```